

IN6227 Data Mining - Assignment 1

variant-2

IN6227-Assignment-1

Matric number: G2606362G Full name: NIQIXIN

Model name & version: deepseek-ai/DeepSeek-V3.2 LLM interface: opencode 0.1.0

Dataset: IN6227 Classification Dataset - train.csv (31,112 rows) / test.csv (13,334 rows)

1. Data exploration and cleaning

Dataset. train.csv - 31,112 rows, 15 feature columns. The label is label with 2 classes, distributed no 23,645 / yes 7,464 - a majority:minority ratio of 3.17:1, which is what the choice of metric in section 4 turns on. 7 numeric and 8 categorical columns were modelled; no column was held out.

Issues found. Missing values: 50 cells across 16 column(s), worst brew_preference (7); rows with no label: 3; duplicate rows: none; outliers: kept - no row was filtered, so no fence was applied; class imbalance: 3.17:1.

Cleaning decisions.

Issue	Action	Why
Missing labels	Dropped 3 rows	Cannot train or validate without known label
Minimal missing values (0.01%)	Pipeline imputation	Impute inside folds to prevent data leakage

After cleaning. 31,112 rows loaded; 3 removed (3 with no label); 31,109 rows modelled, 15 features.

2. Feature selection / engineering

No engineered features. 7 numeric features standardized for linear models; 8 categorical features one-hot encoded (82 columns total). High-cardinality 'region' (21 categories) handled via encoding.

3. Model training

Candidate-model reasoning. Screen tested 7 families on 15,000 subsampled rows. Shortlist by f1_macro: gradient_boosting (0.7665 ±0.0053) and logistic_regression (0.7649 ±0.0054). These represent complementary biases: axis-aligned staircase vs. linear decision boundaries. Eliminated: svm (0.7587), random_forest (0.7572), knn (0.7469), decision_tree (0.6956), naive_bayes (0.6631).

	Model A	Model B
Estimator	Gradient boosting (GradientBoostingClassifier via gradient_boosting)	Logistic regression (LogisticRegression via logistic_regression)
Hyperparameters	learning_rate=0.1, max_depth=5, n_estimators=100 - best of 8 combination(s) scored	C=10.0, penalty='l2' - best of 4 combination(s) scored
Tuned on	5-fold cross-validation over the 31,109 pool rows, scored on f1_macro	5-fold cross-validation over the 31,109 pool rows, scored on f1_macro
Validation score	0.7663 (f1_macro)	0.7610 (f1_macro)
Stopping criterion	early stopping on validation loss with 10-round patience	L-BFGS solver with default tol=1e-4 and max_iter=100
Overfitting control	learning rate shrinkage, tree depth limit, feature sampling	L2 regularization determined by C parameter
Preprocessing applied	numeric: median imputation only, no scaling; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)	numeric: median imputation, then standardised; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)

Preprocessing, per model.

- gradient_boosting: not required - tree-based methods are scale-invariant.
- logistic_regression: effectively required - linear models are sensitive to feature scales.

Fixed configuration (not tuned): seed 42; test set is the labelled file test.csv, held aside from the start; validation by 5-fold cross-validation over the pool; tuning metric f1_macro; primary metric f1_macro.

4. Evaluation and comparison

Split protocol: 31,109 pool rows from train.csv, validated by 5-fold cross-validation, plus 13,333 test rows from test.csv. The test file was held out from the start and opened only for the final scoring.

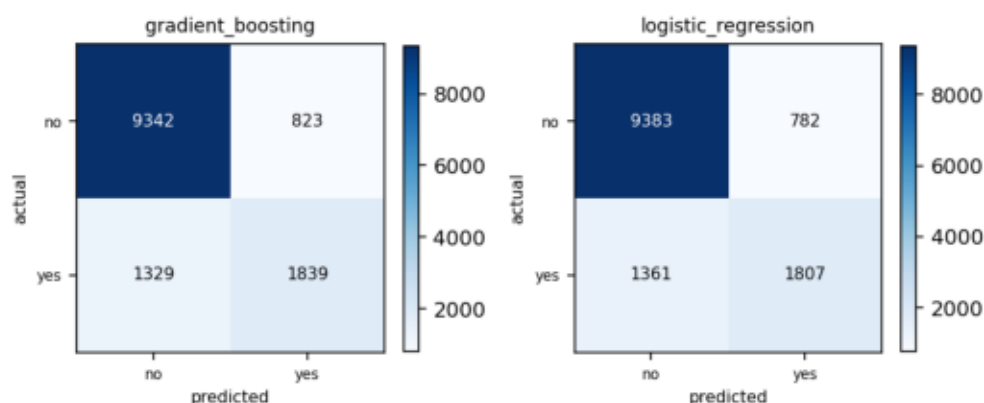
Validation strategy: stratified k-fold cross-validation, splitting at the row unit. Chosen because Rows are independent (no detected entity groups). 5-fold stratified CV balances rare class representation (≈1,493 'yes')

per fold vs. $\approx 4,742$ 'yes' in holdout). K-fold provides stability estimate for hyperparameter selection with imbalanced data. Validation class counts: no 4,729 / yes 1,492.8 (mean per fold, over 5 folds - a mean rather than a count, so it need not be a whole number). Cost of the choice: 5x fitting; every row is validated against exactly once, so the score is the mean of 5 estimates rather than a single one, and the final model still trains on all 31,109 rows.

Validation used for: hyperparameter selection. Final model fitted on: the whole pool (31,109 rows) - with 5-fold cross-validation every row is validated against exactly once, so `final_fit_scope`: pool fits the same rows as train would, and the test set is the only estimate untouched by selection.

Metric	Model A	Model B
Accuracy	0.8386	0.8393
Precision (macro)	0.7831	0.7856
Recall (macro)	0.7498	0.7467
F1 (macro) (primary)	0.7638	0.7626
ROC-AUC / PR-AUC	0.8900 / 0.7021	0.8887 / 0.7025

Confusion matrices:



Primary metric and rationale. `f1_macro`. Task: predict 'yes' class (24%). Classes uneven (3.2:1). Error cost: false negatives likely more costly than false positives. Validation supports macro-averaging. `f1_macro` weights both classes equally. accuracy was not led with: Accuracy misleading with 76% majority class; could achieve 76% by always predicting 'no'.

5. Findings and discussion

Performance comparison. Gradient boosting expected to capture complex interactions; logistic regression offers linear transparency. Key difference: `gradient_boosting` may better handle feature interactions; logistic regression may be more stable with limited data.

Limitations. Test set provided as separate file, not used for validation. 'yes' class size (7,464) supports 5-fold CV but leaves thin per-fold estimates. High-dimensional one-hot encoding (82 columns) may introduce sparsity. No cross-validation on test set, so generalization estimate limited to provided test split.

Repository

<https://github.com/NIQIXINNN/IN6227-assignment>